

Steady-state conduction in heat transfer

Fins

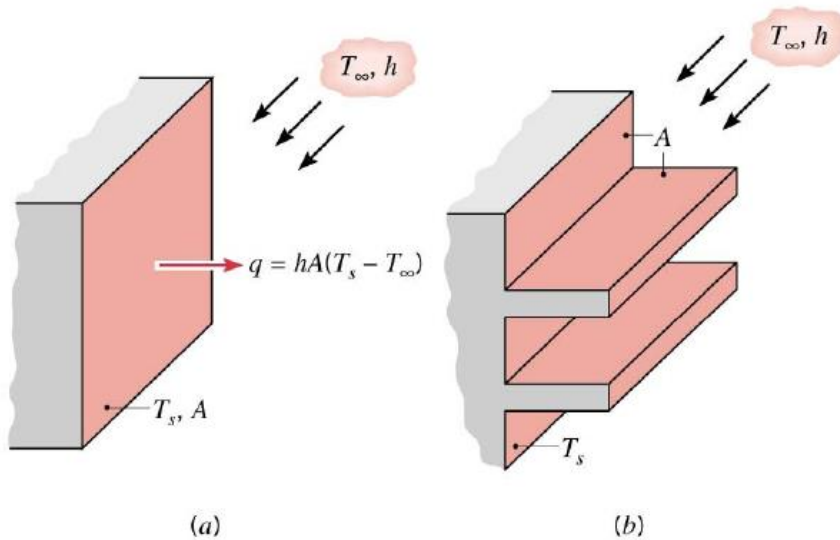


FIGURE 3.12 Use of fins to enhance heat transfer from a plane wall. (a) Bare surface. (b) Finned surface.

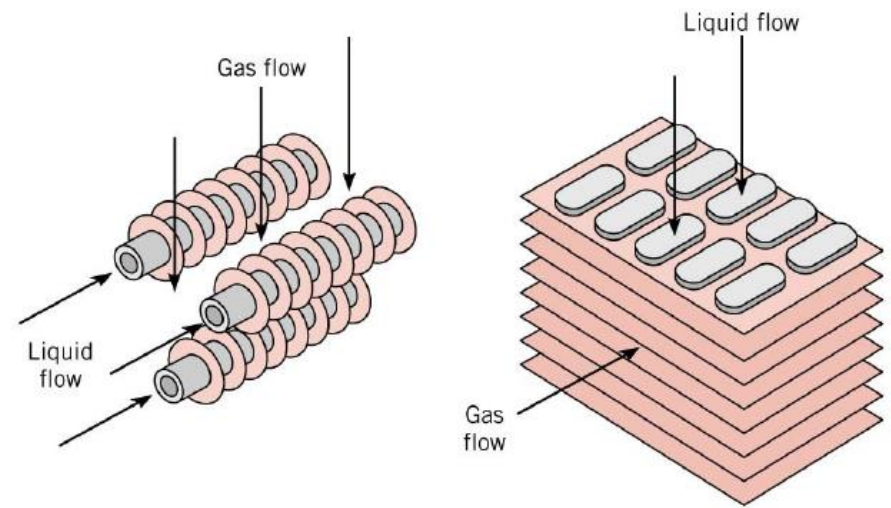
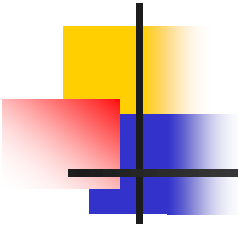


FIGURE 3.13 Schematic of typical finned-tube heat exchangers.



Heat sink

- ❑ 散熱鰭片 (Fins of heat sink)
- ❑ A **heat sink** (or **heatsink**) is an environment or object that absorbs and dissipates heat from another object using thermal contact (either direct or radiant). Heat sinks are used in a wide range of applications wherever efficient heat dissipation is required; major examples include refrigeration, heat engines, cooling electronic devices and lasers.
(http://en.wikipedia.org/wiki/Heat_sink)

Heat sink (cont.)

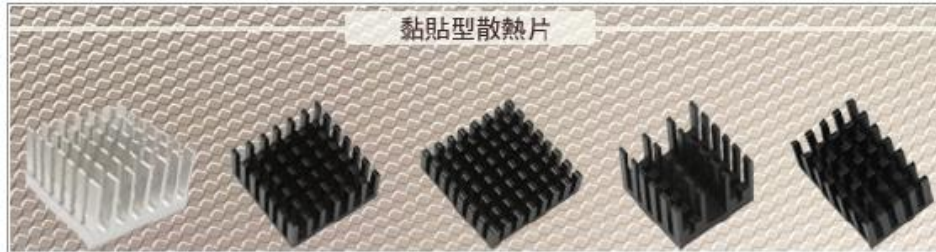


柏旭佳國際企業有限公司

<http://www.pro-best.com.tw/index.php?cPath=70>

Heat sink (cont.)

散熱片



黏貼型散熱片

- 機殼內設定環境溫度為55℃，並選用相近之晶體尺寸測試
- 散熱片皆貼上 CTD-01-S8 導熱背膠

參考可 降溫度 型號 (長*寬*高)	晶體 功率	1 W	2 W	3 W	4 W	5 W	6 W	7 W	8 W	9 W	10 W
CT-S160160 (16 x 16 x 12 mm)		8	10	13	15	x	x	x	x	x	x
CT-S220140 (22 x 14 x 9 mm)		8	11	14	17	x	x	x	x	x	x
CT-S220150 (22 x 15 x 11.7 mm)		10	14	18	22	26	x	x	x	x	x
CT-S220229 (22 x 22 x 9 mm)		10	13	16	19	x	x	x	x	x	x
CT-S220221 (22 x 22 x 11.7 mm)		5	7	9	11	13	15	x	x	x	x
CT-S279279 (27.9x27.9x11.4 mm)		6	9	12	15	18	21	24	x	x	x
CT-S280280 (28x 28 x 8 mm)		5	7	9	11	13	15	x	x	x	x
CT-S350356 (35 x 35 x 6 mm)		4	6	8	10	12	14	x	x	x	x
CT-S350358 (35 x 35x 8.5 mm)		6	8	10	12	14	16	18	x	x	x


陸采企業有限公司
 Centrans Enterprise Co.,Ltd.

http://www.centran.com.tw/cooler_heatsink.html

Heat sink (cont.)



CPU Cooler for Intel P4 Solution

CEP429151D

Intel FMB2 Thermal Solution

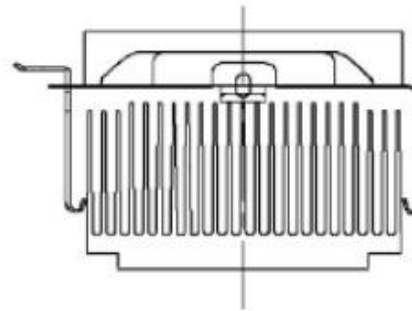
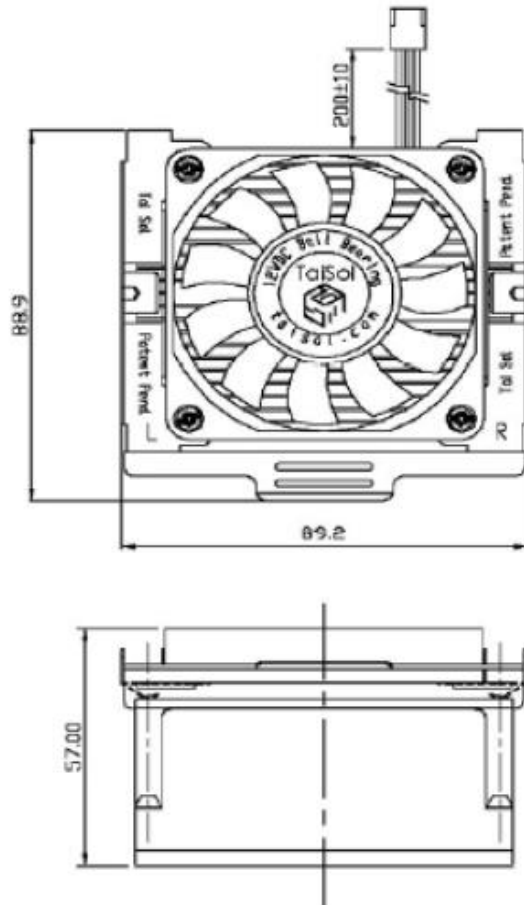
- Support 2.8Ghz CPU and beyond
- Passed 3rd party test house requirements for Thermal
- 50G shock and vibration test
- Dow Corning DC340 thermal interface grease
- Proper amount of contact force on the CPU die
- Patent pending lockable level clip design makes assembly and removal a breeze !
- Retention module and back Plate not included



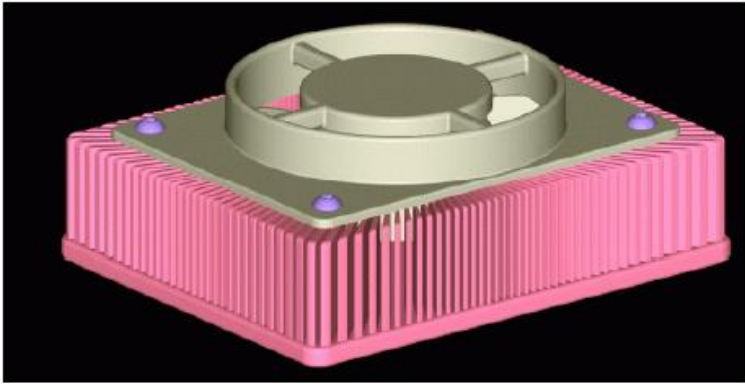
泰碩電子

<http://www.taisol.com.tw/index.htm>

Heat sink (cont.)



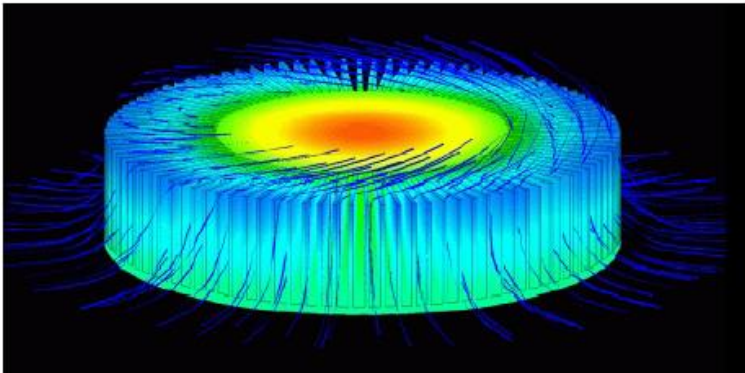
Heat sink (cont.)



CAST ALUMINUM HEAT SINK

Developed for a leading semiconductor company, this low-cost cast aluminum heat sink measures 89mm by 64mm by 35mm (including blower), with a total thermal resistance of 0.29°C/W .

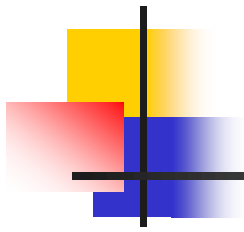
Total weight 195 grams.



Microelectronic Thermal Management

Novel Concepts, Inc.

<http://www.novelconceptsinc.com/heat-sinks.htm>



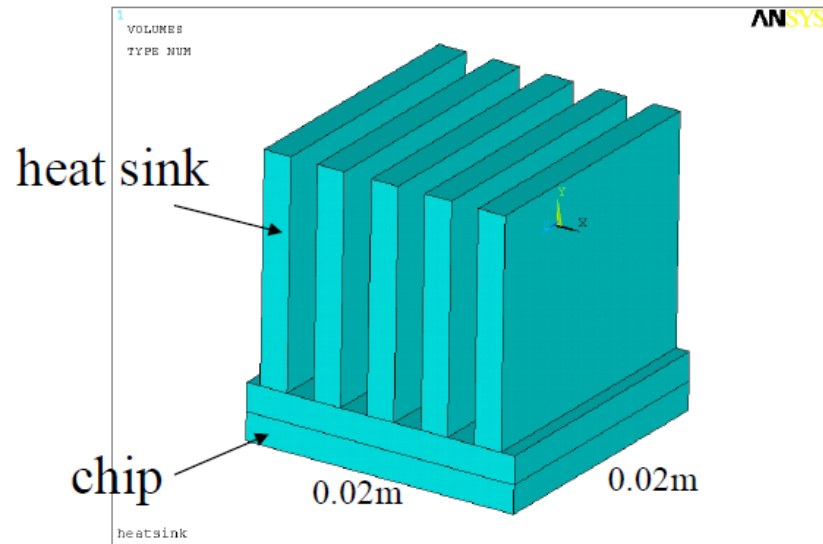
Example 5

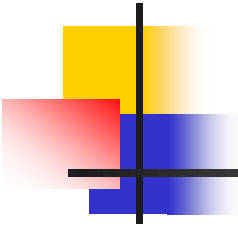
題目

- ❑ 散熱鰭片 (Fins of heat sink), 鋁製 (擠製), $k = 237 \text{ W/m}^\circ\text{C}$
- ❑ 晶片 (chip), 發熱 6 W , $k = 150 \text{ W/m}^\circ\text{C}$
- ❑ 對流: $h = 50 \text{ W/m}^2^\circ\text{C}$, $T_\infty = 25^\circ\text{C}$
- ❑ 三維穩態熱傳導分析
- ❑ 求晶片溫度

$$\frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + \dot{q} = 0$$

$$\dot{q} = 6 \text{ W/volume}$$



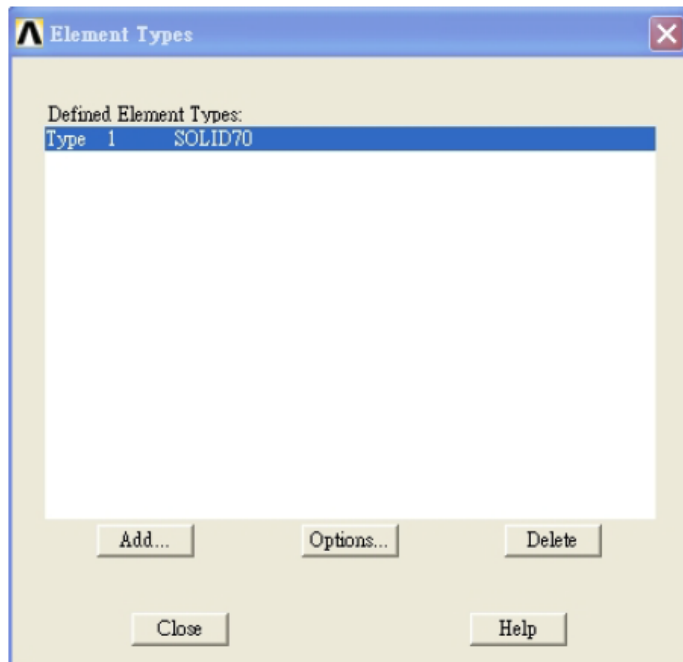


預先設定

- ☐ Jobname: heatsink
- ☐ Title: heatsink
- ☐ Preference: Thermal

元素

□ SOLID 70

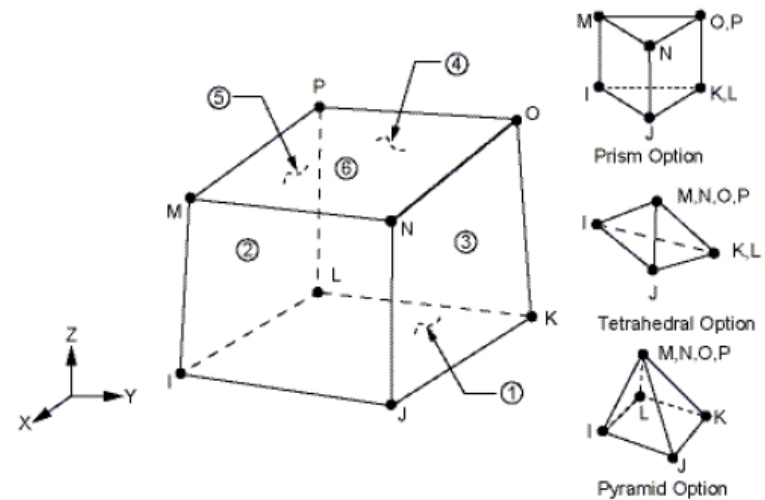


ANSYS Release 10.0 Documentation for ANSYS

[Element Reference](#) | [Part I. Element Library](#) |

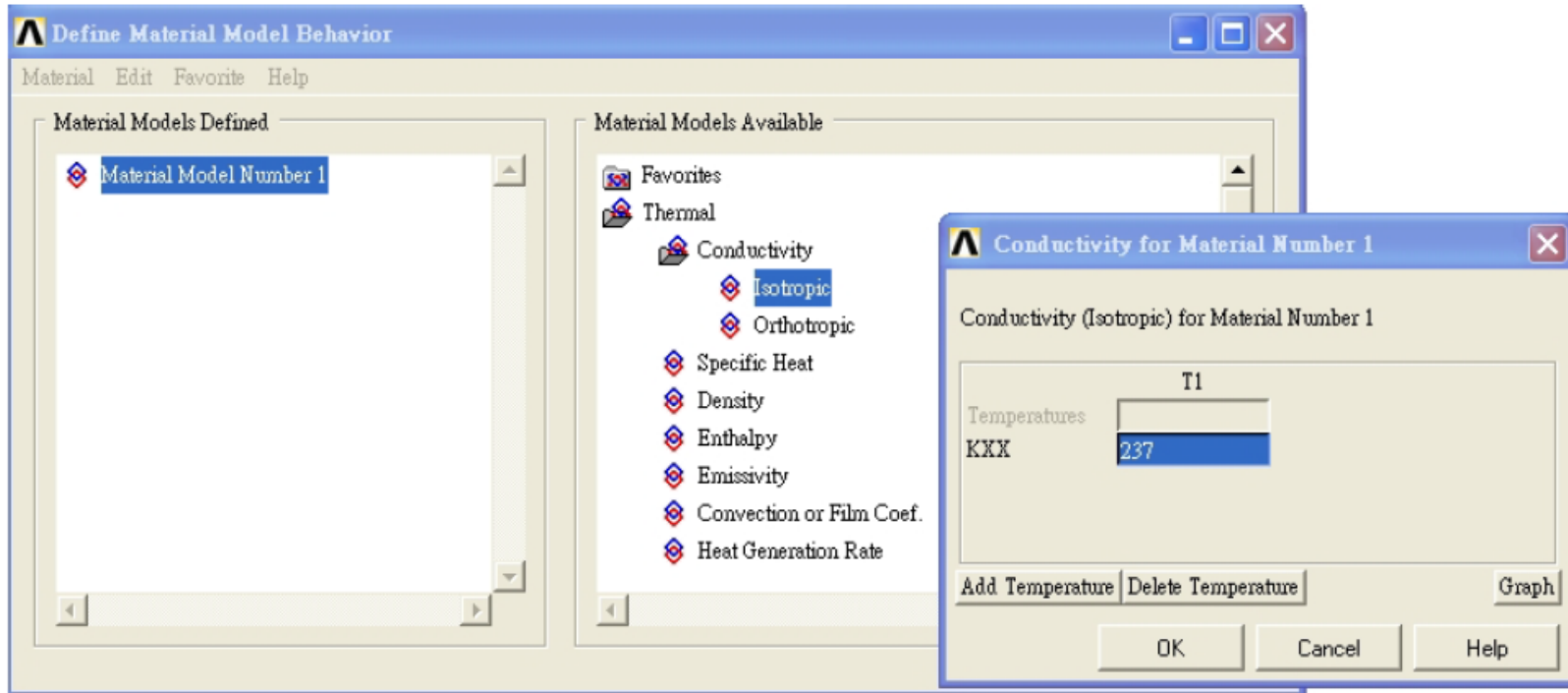
SOLID 70 3-D Thermal Solid

Figure 70.1 SOLID70 Geometry



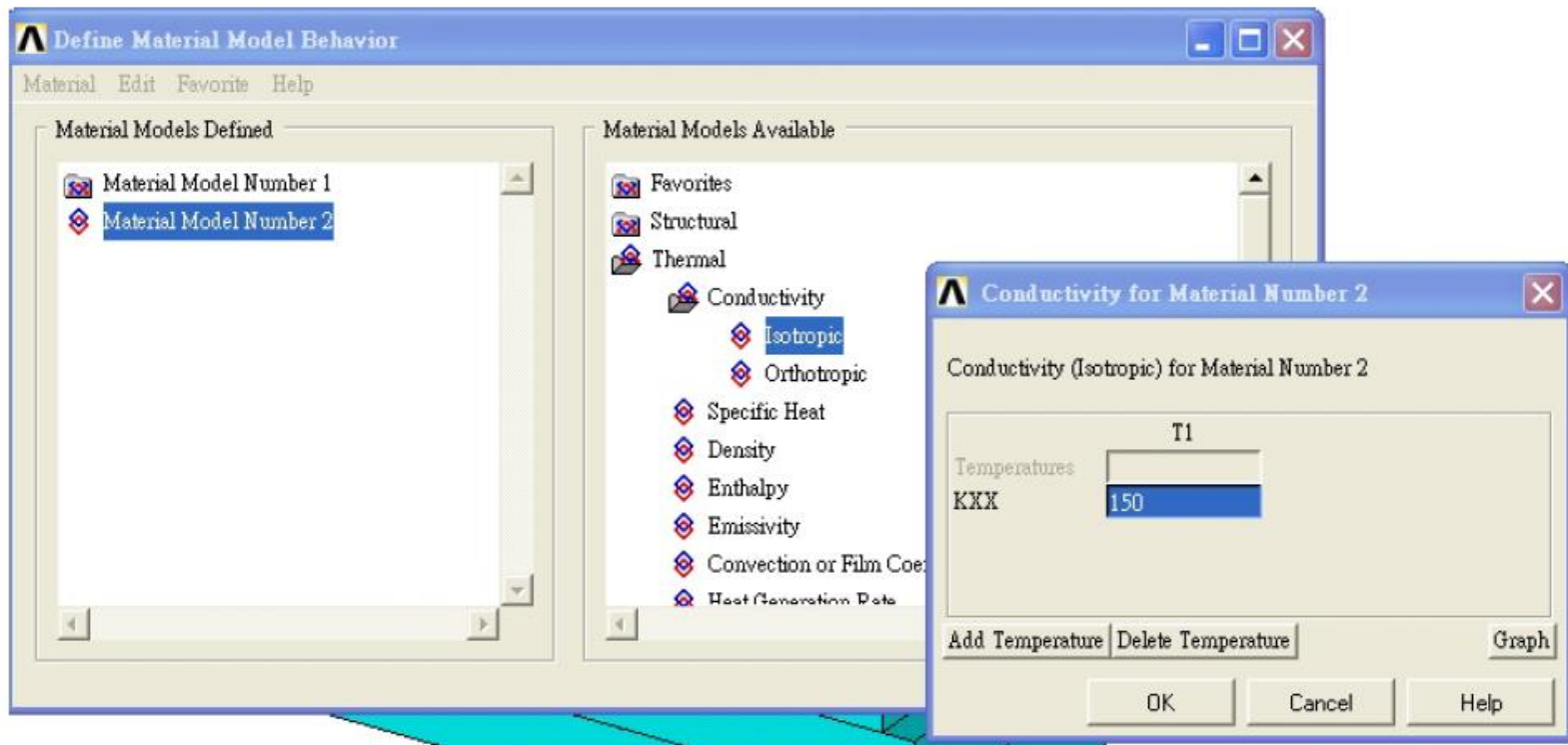
材料係數

$k = 237 \text{ W/m}^\circ\text{C}$ (MAT 1)

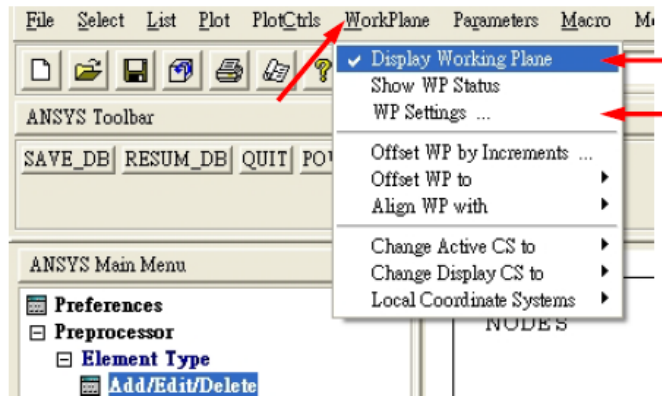


材料係數(續)

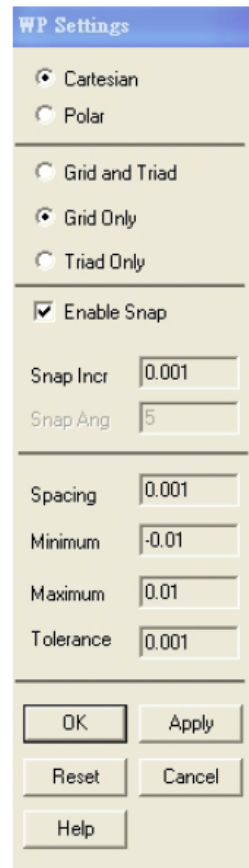
$k = 150 \text{ W/m}^\circ\text{C}$ (MAT 2)



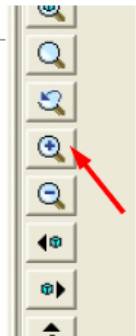
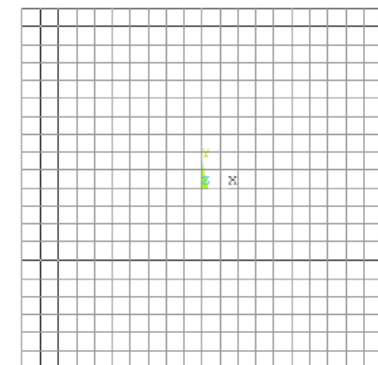
幾何



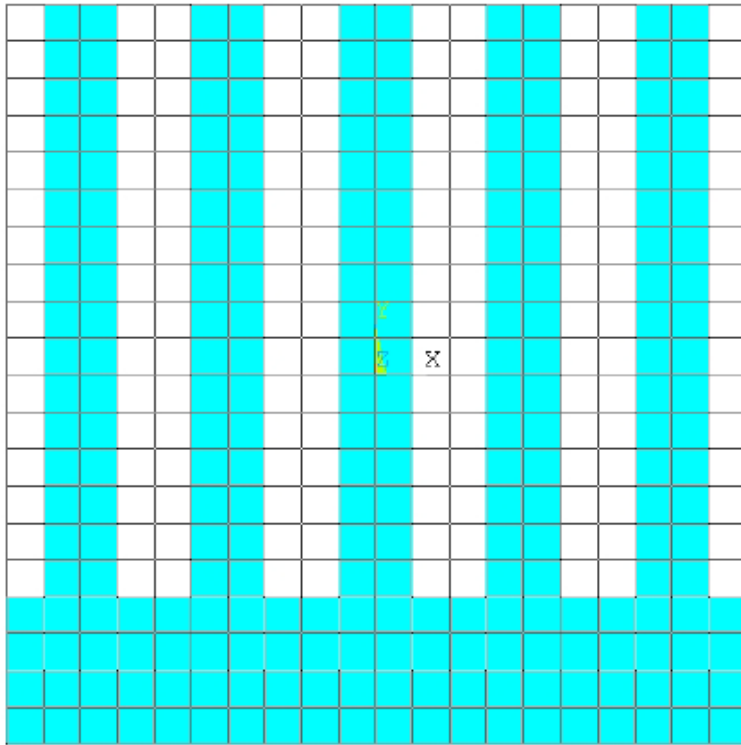
單位: m →



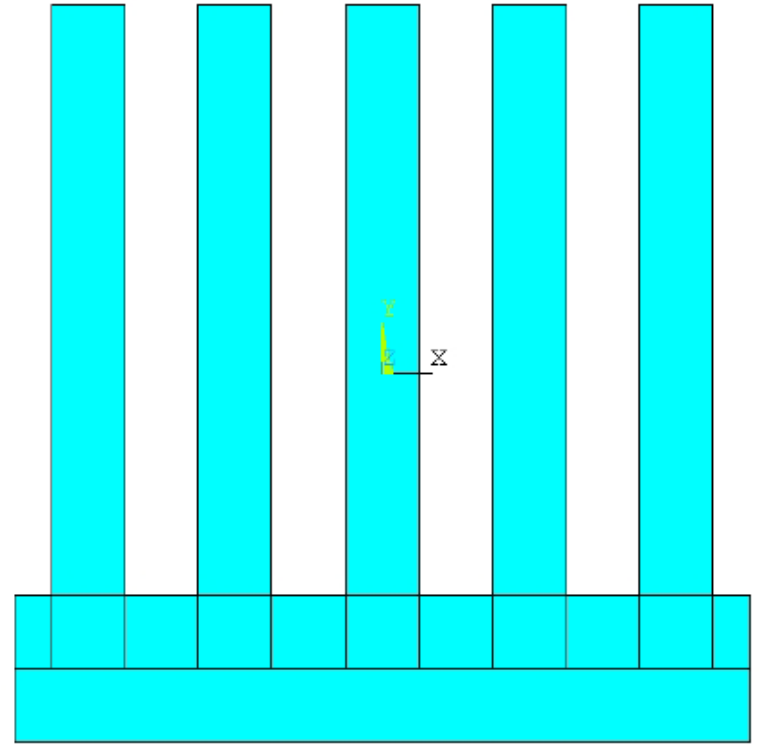
Working plane



幾何(續)

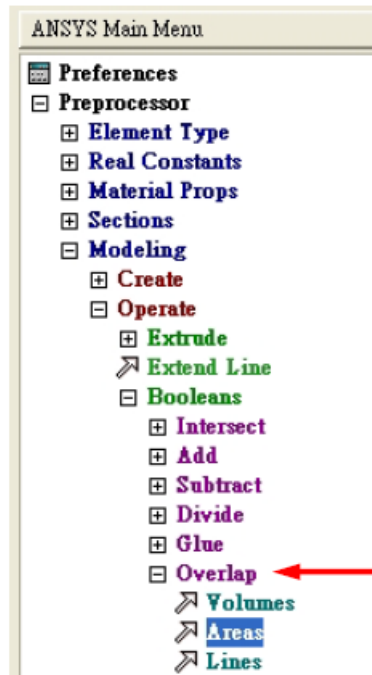


0.02m

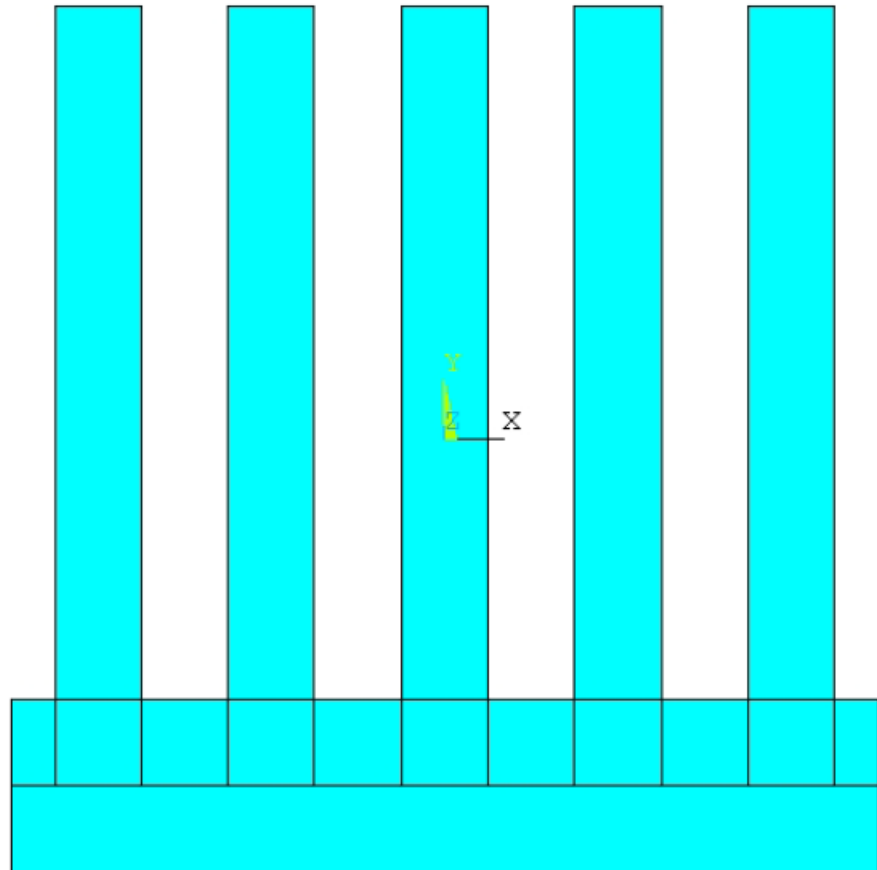


不顯示 working plane

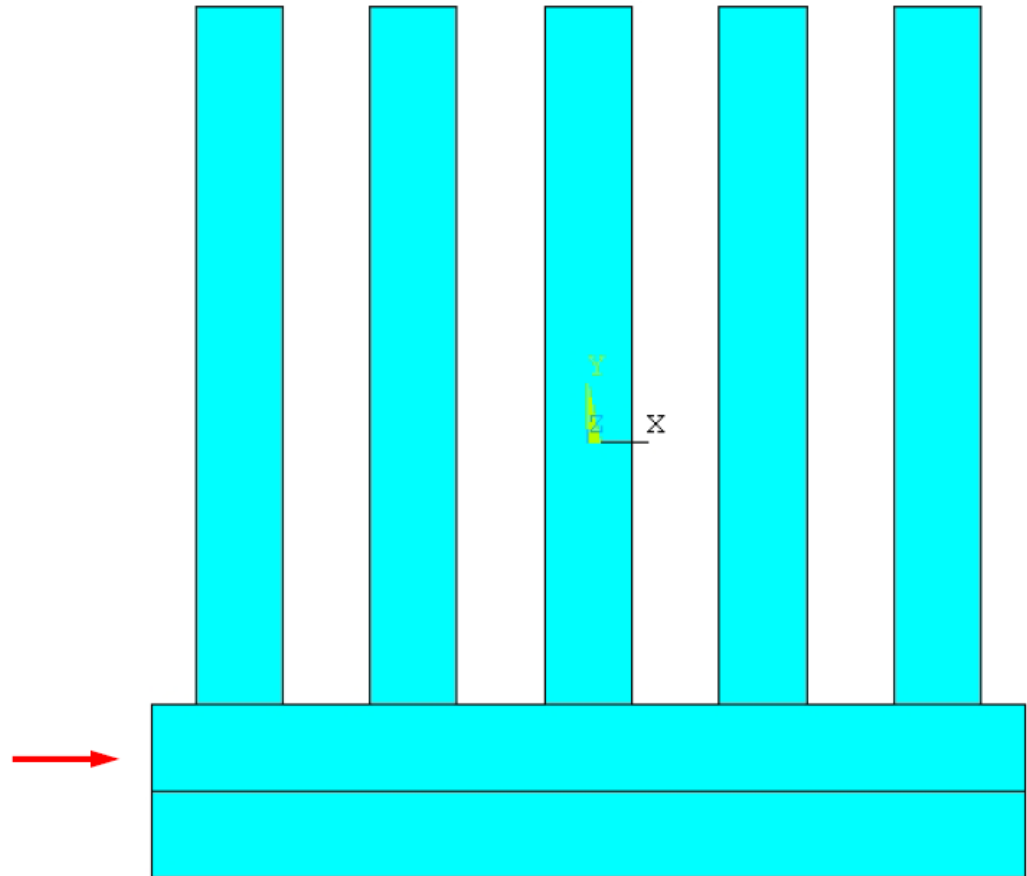
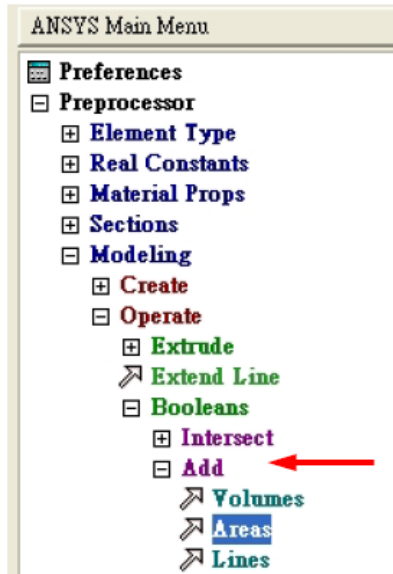
幾何(續)



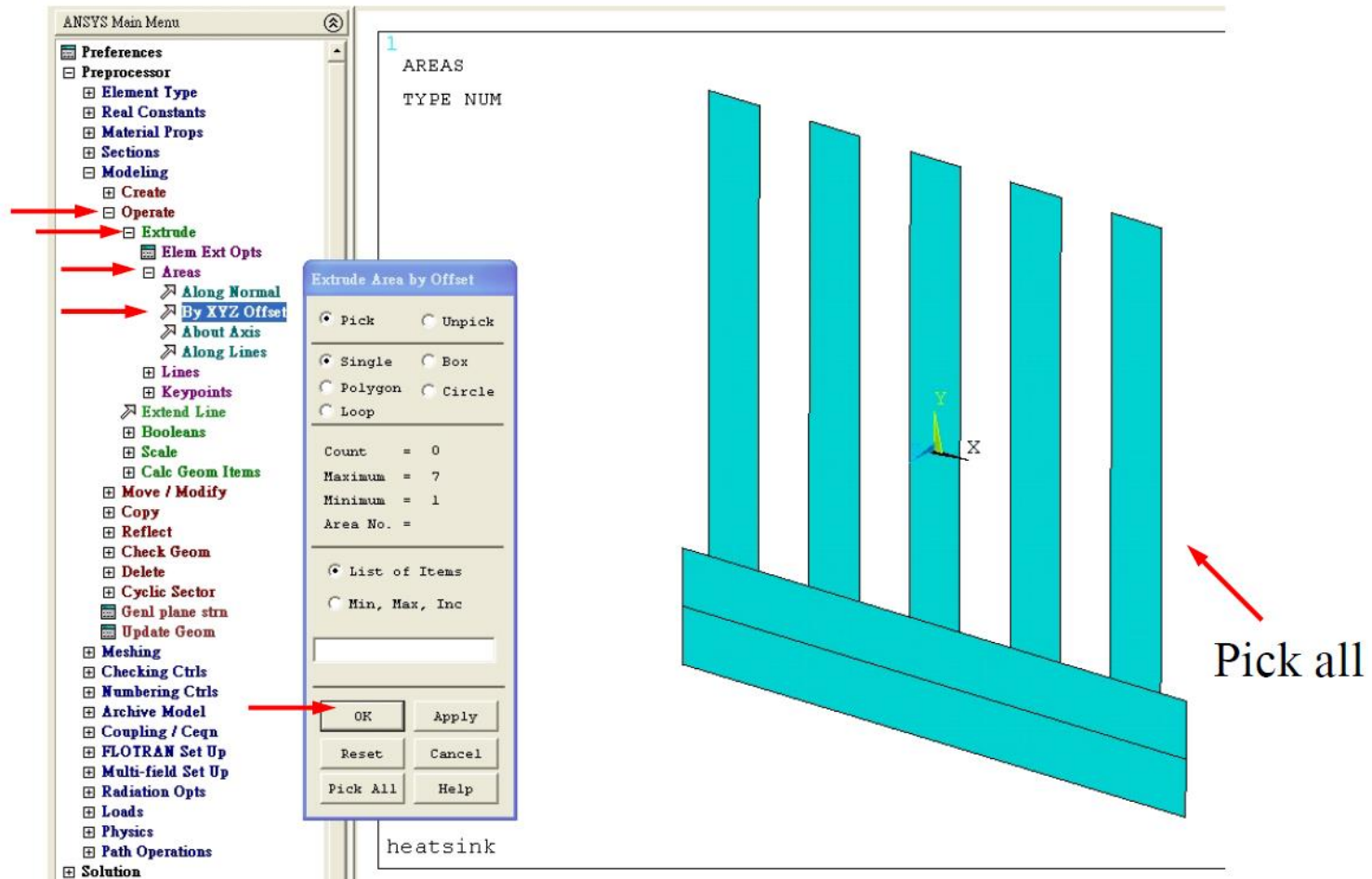
Pick all



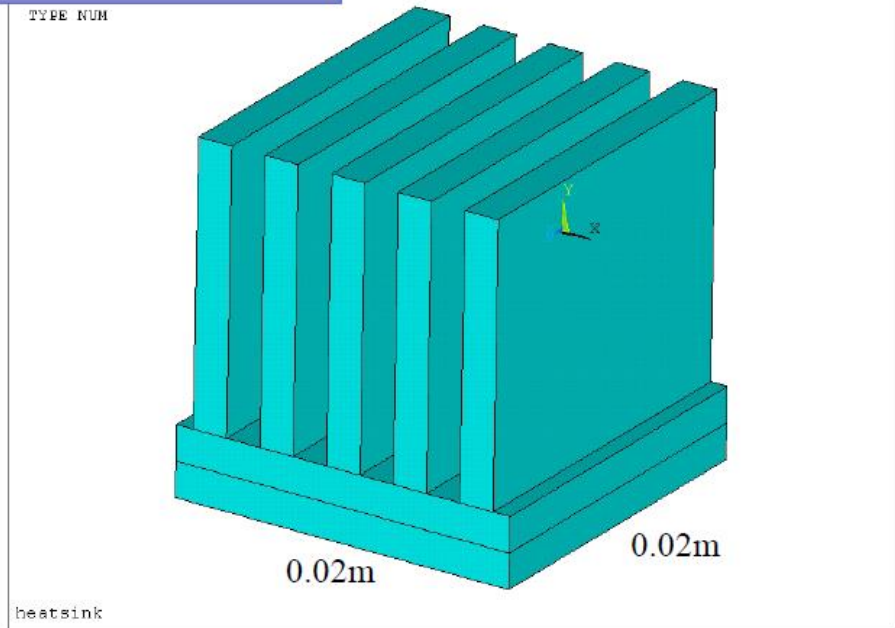
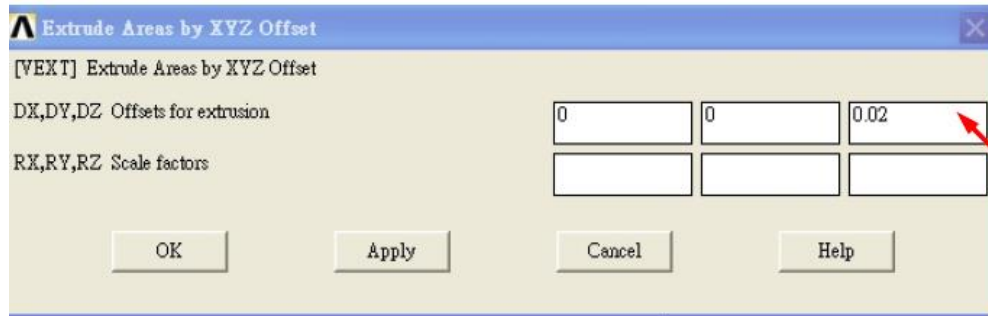
幾何(續)



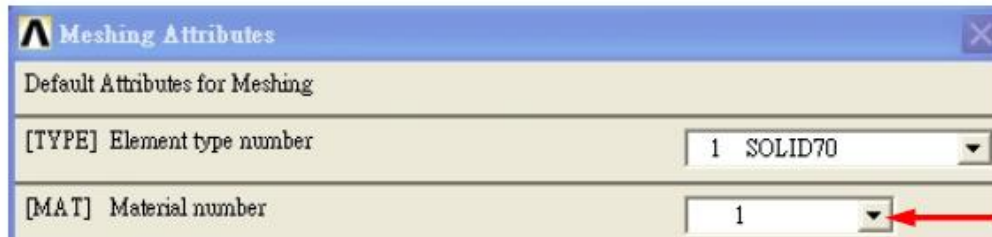
幾何(續)



幾何(續)



Mesh



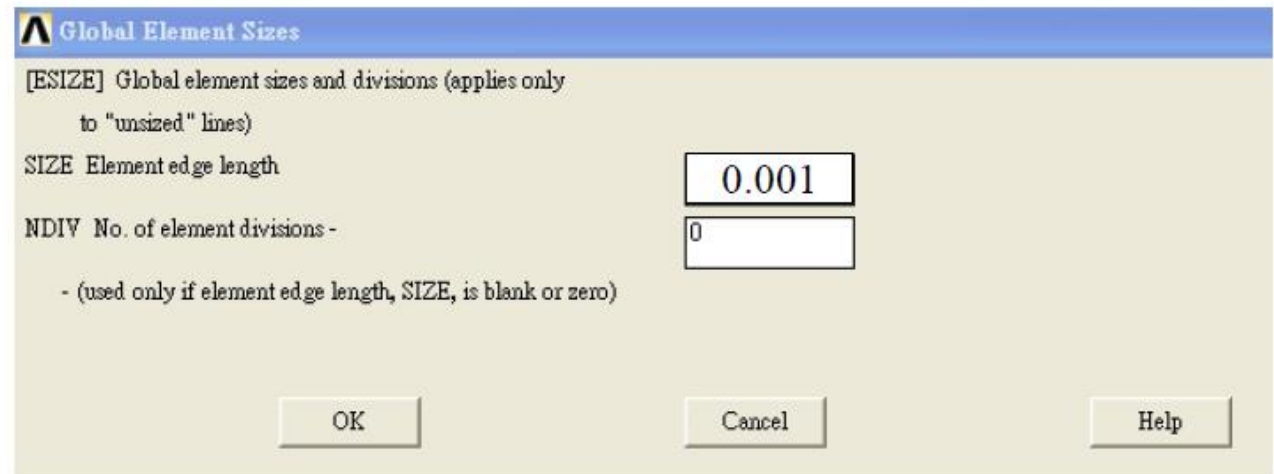
Meshing Attributes

Default Attributes for Meshing

[TYPE] Element type number: 1 SOLID70

[MAT] Material number: 1

A red arrow points to the dropdown arrow of the [MAT] Material number field.



Global Element Sizes

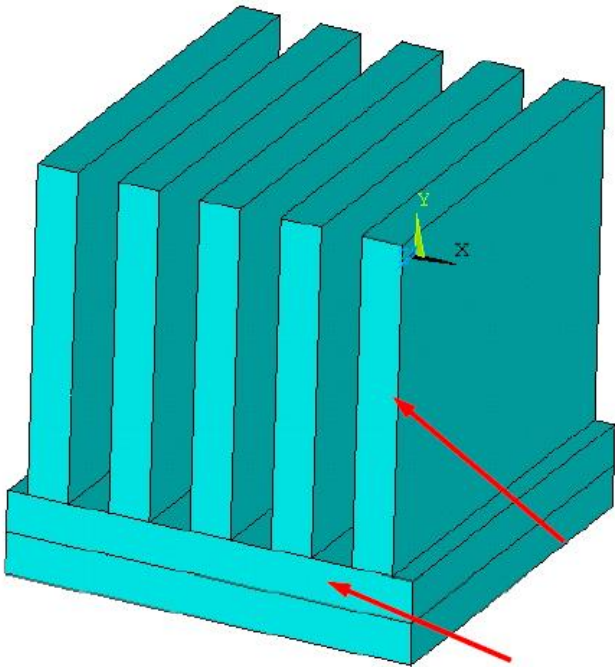
[ESIZE] Global element sizes and divisions (applies only to "unsized" lines)

SIZE Element edge length: 0.001

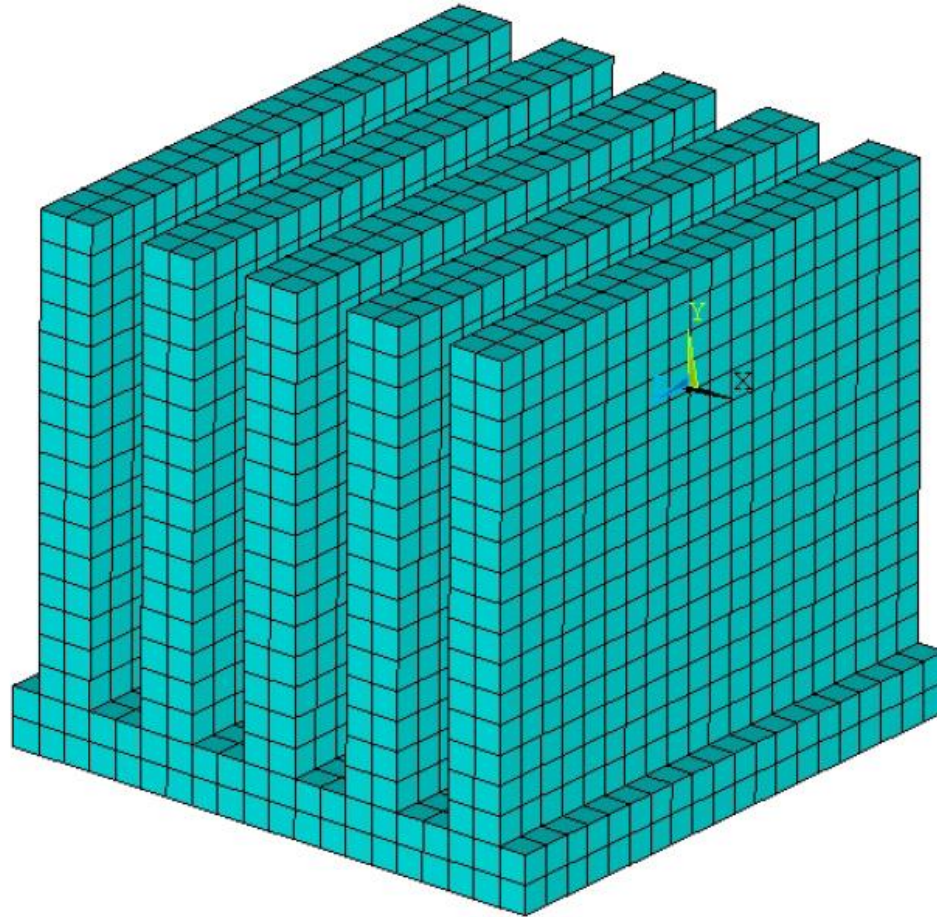
NDIV No. of element divisions: 0

- (used only if element edge length, SIZE, is blank or zero)

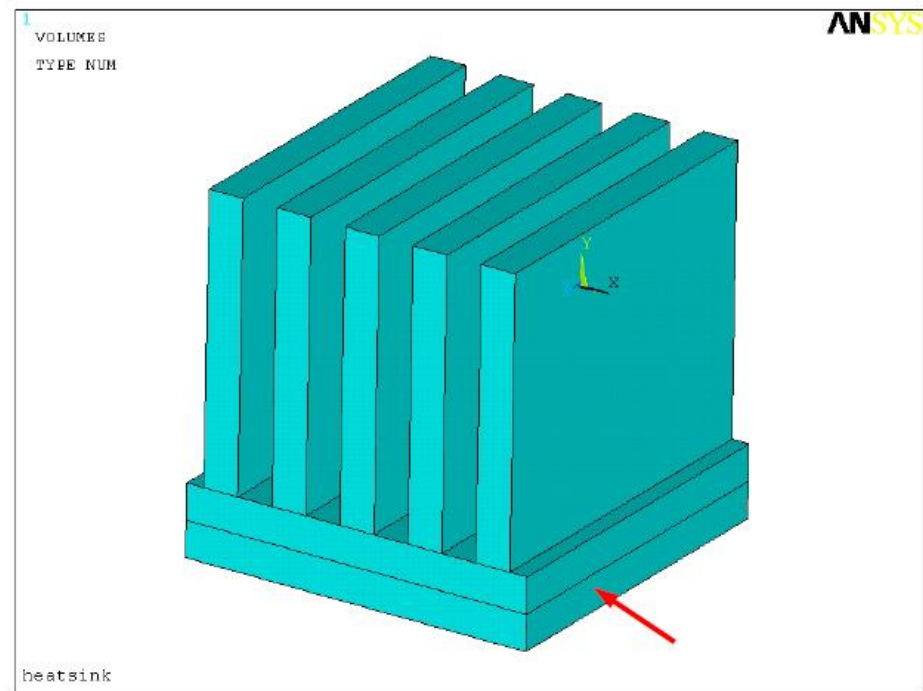
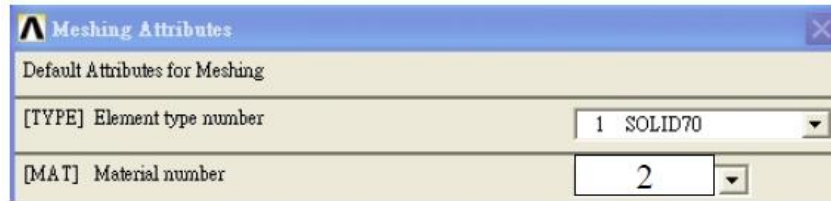
OK Cancel Help



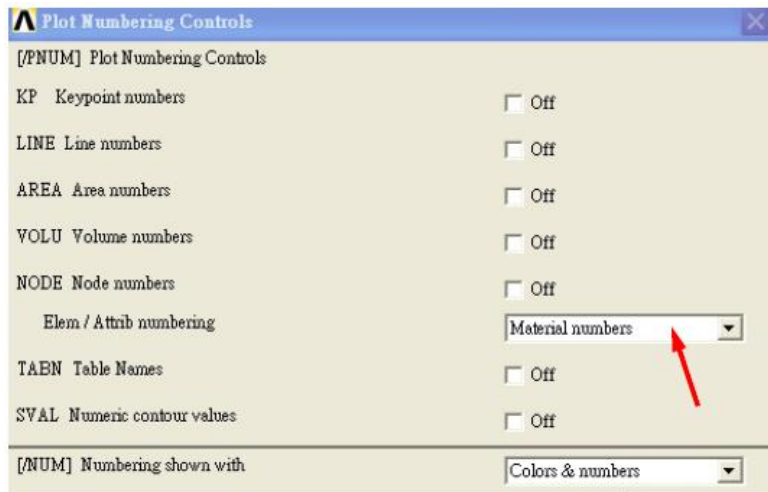
Mesh (續)



Mesh (續)



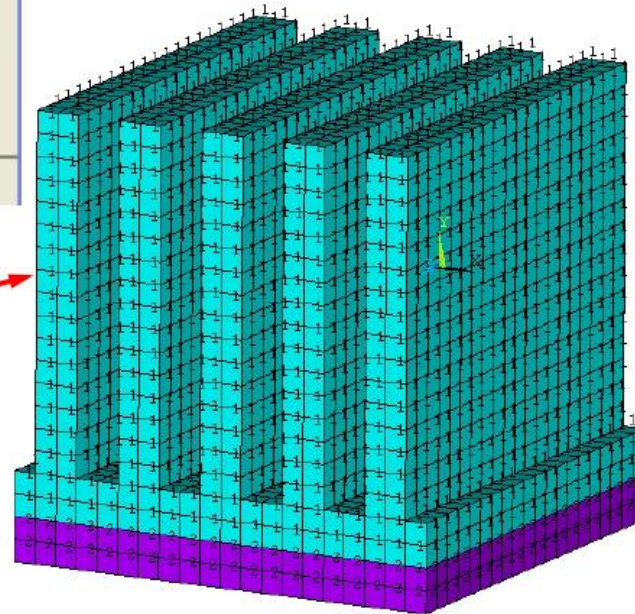
Mesh (續)



heatsink
(MAT 1)

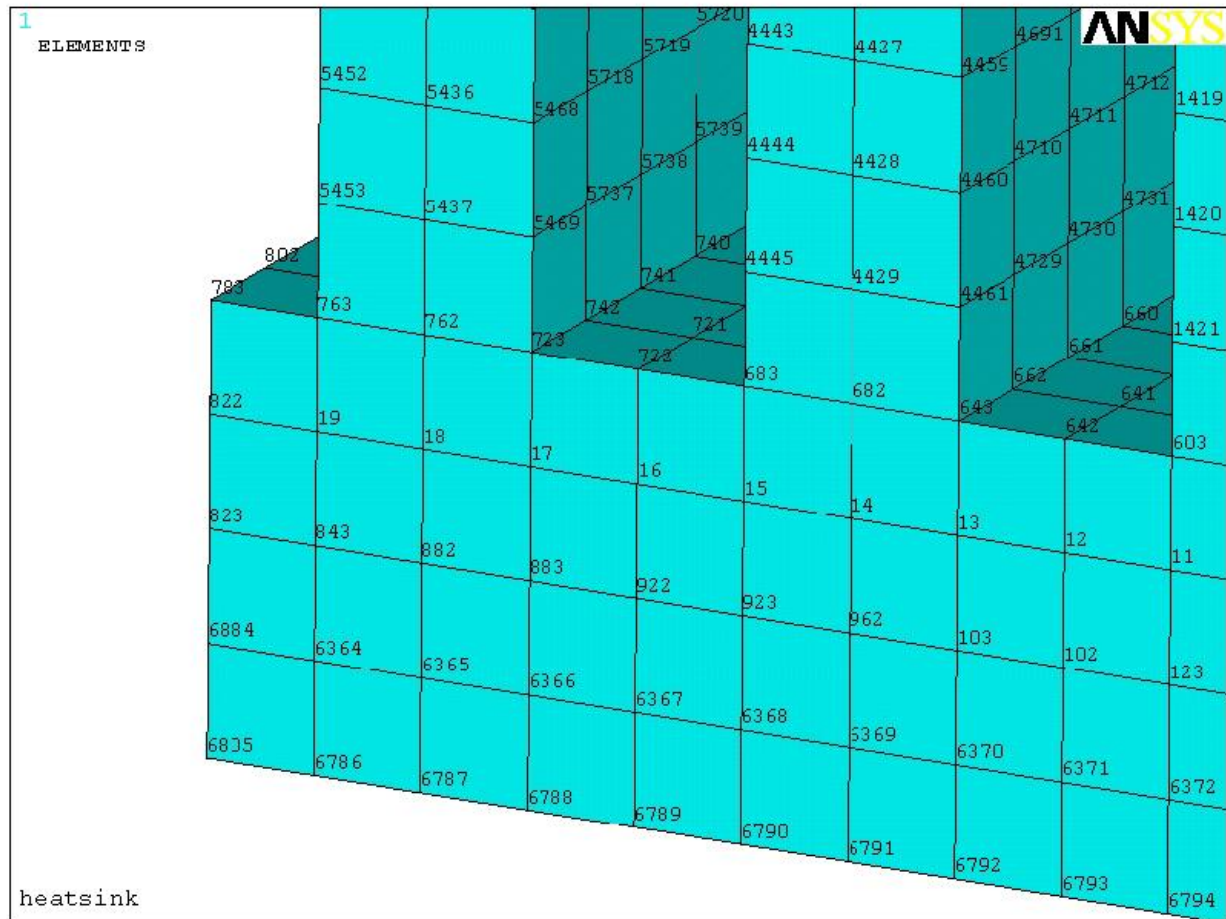
chip
(MAT 2)

heatsink

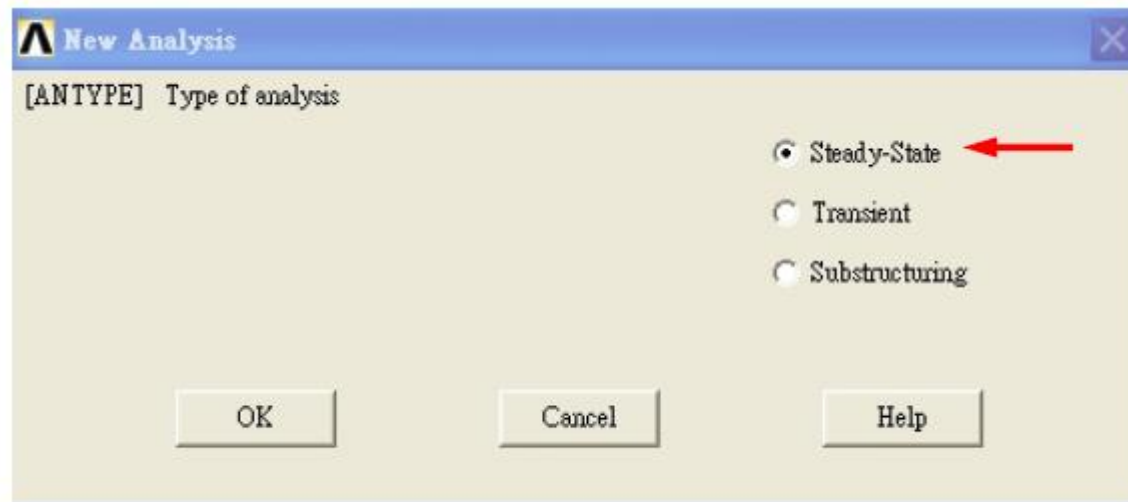


Mesh (續)

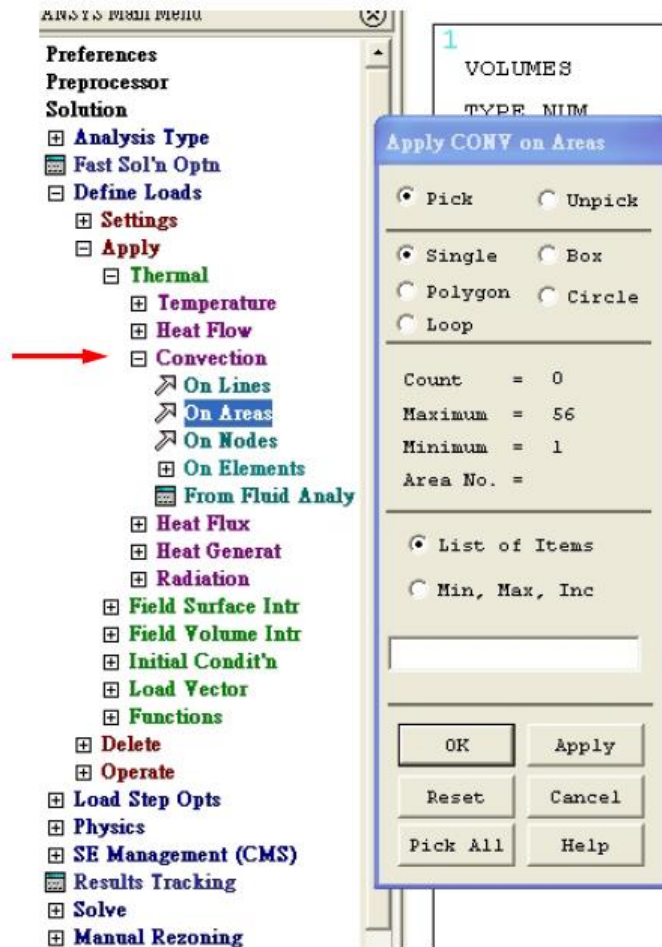
檢查



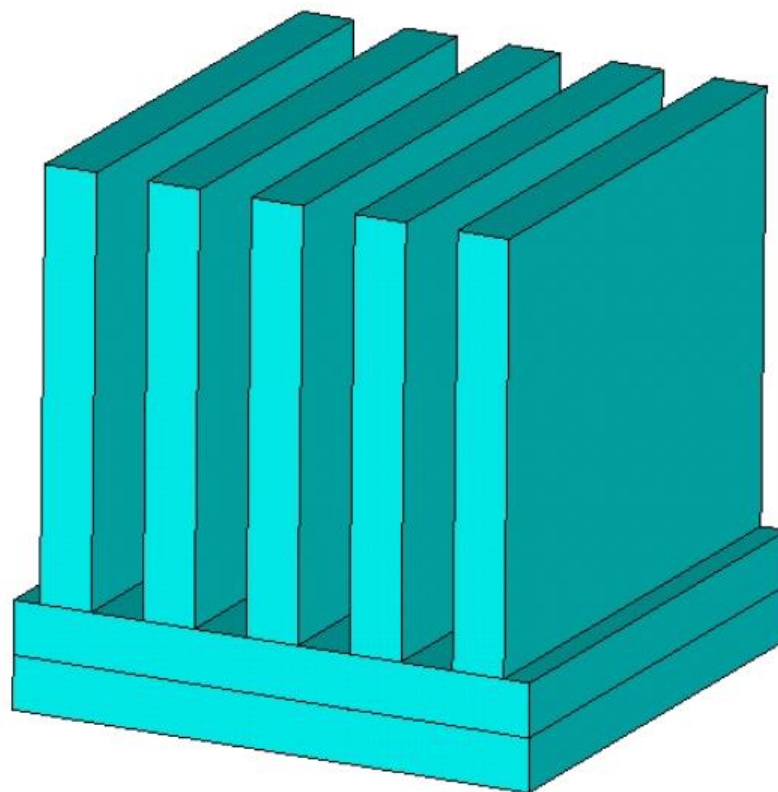
分析型式



邊界條件



pick所有與空氣接觸之面



邊界條件(續)

pick所有與空氣接觸之面

Apply CONV on Areas

☒ Pick ☐ Unpick

☐ Single ☒ Box

☐ Polygon ☐ Circle

☐ Loop

Count = 0

Maximum = 56

Minimum = 1

Area No. =

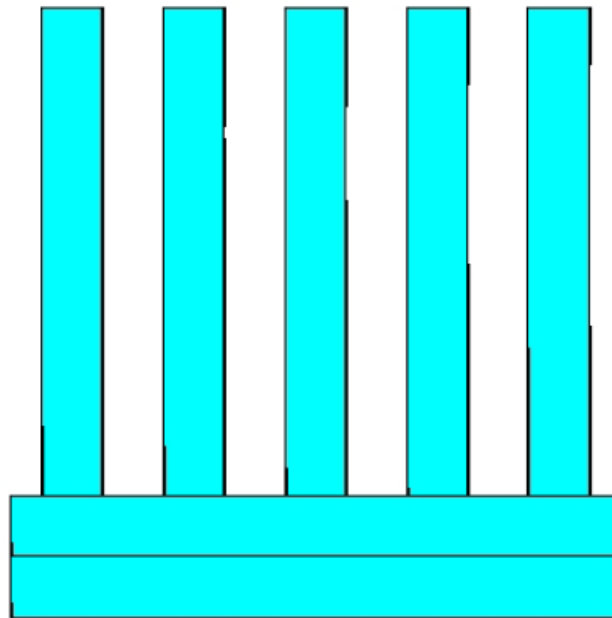
☒ List of Items

☐ Min, Max, Inc

OK Apply

Reset Cancel

Pick All Help



邊界條件(續)

Apply CONV on areas

[SFA] Apply Film Coef on areas Constant value

If Constant value then:
VALI Film coefficient

[SFA] Apply Bulk Temp on areas Constant value

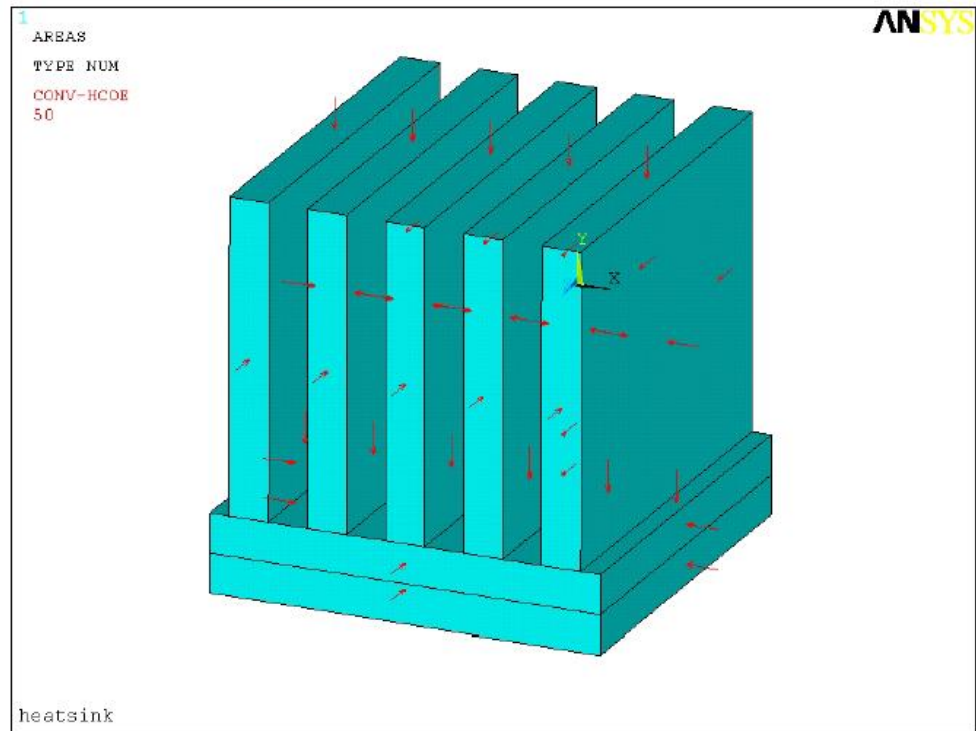
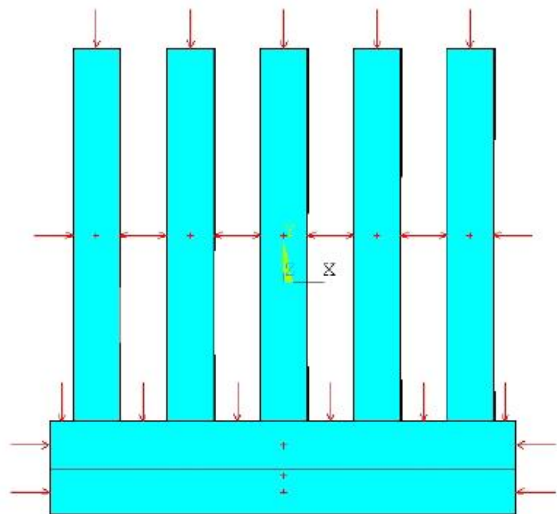
If Constant value then:
VAL2I Bulk temperature

LKEY Load key, usually face no.
(required only for shell elements)

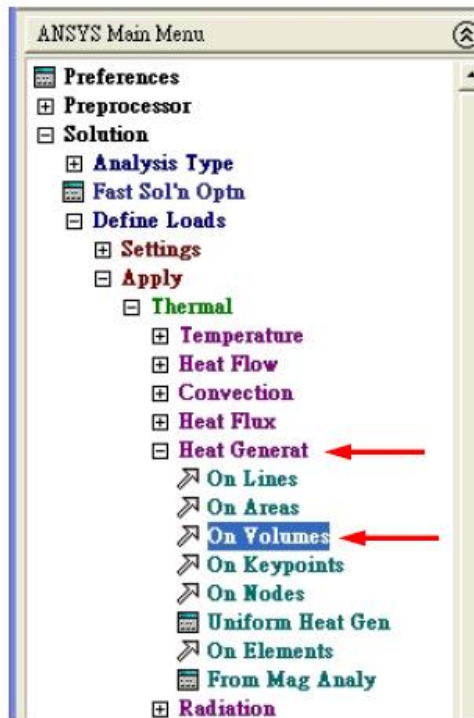
OK Cancel Help

$$h = 50 \text{ W/ m}^2 \text{ }^{\circ}\text{C}, T_{\infty} = 25 \text{ }^{\circ}\text{C}$$

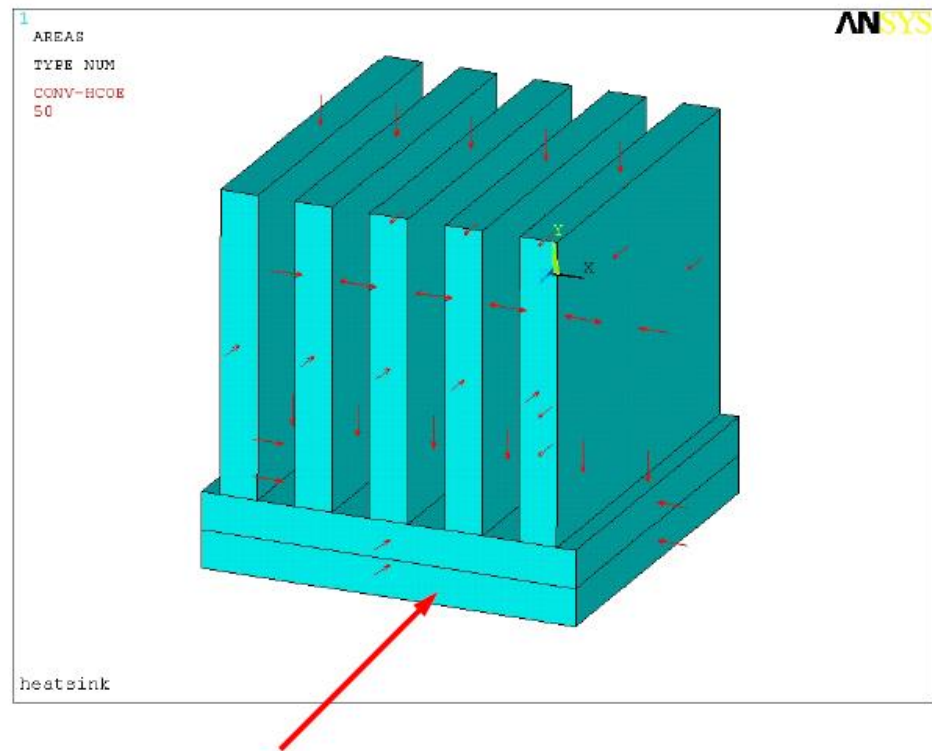
邊界條件(續)



發熱設定



Heat generation on volume



Heat generation on chip volume 晶片(chip), 發熱 6W

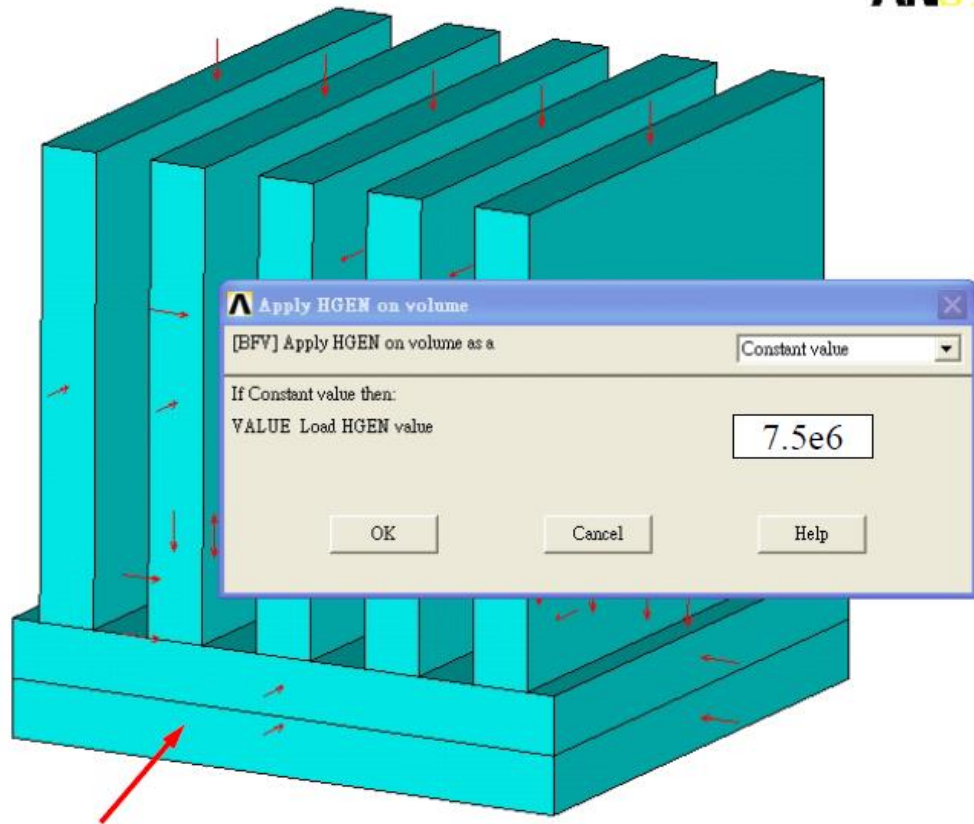
發熱設定(續)

晶片(chip), 發熱 6 W

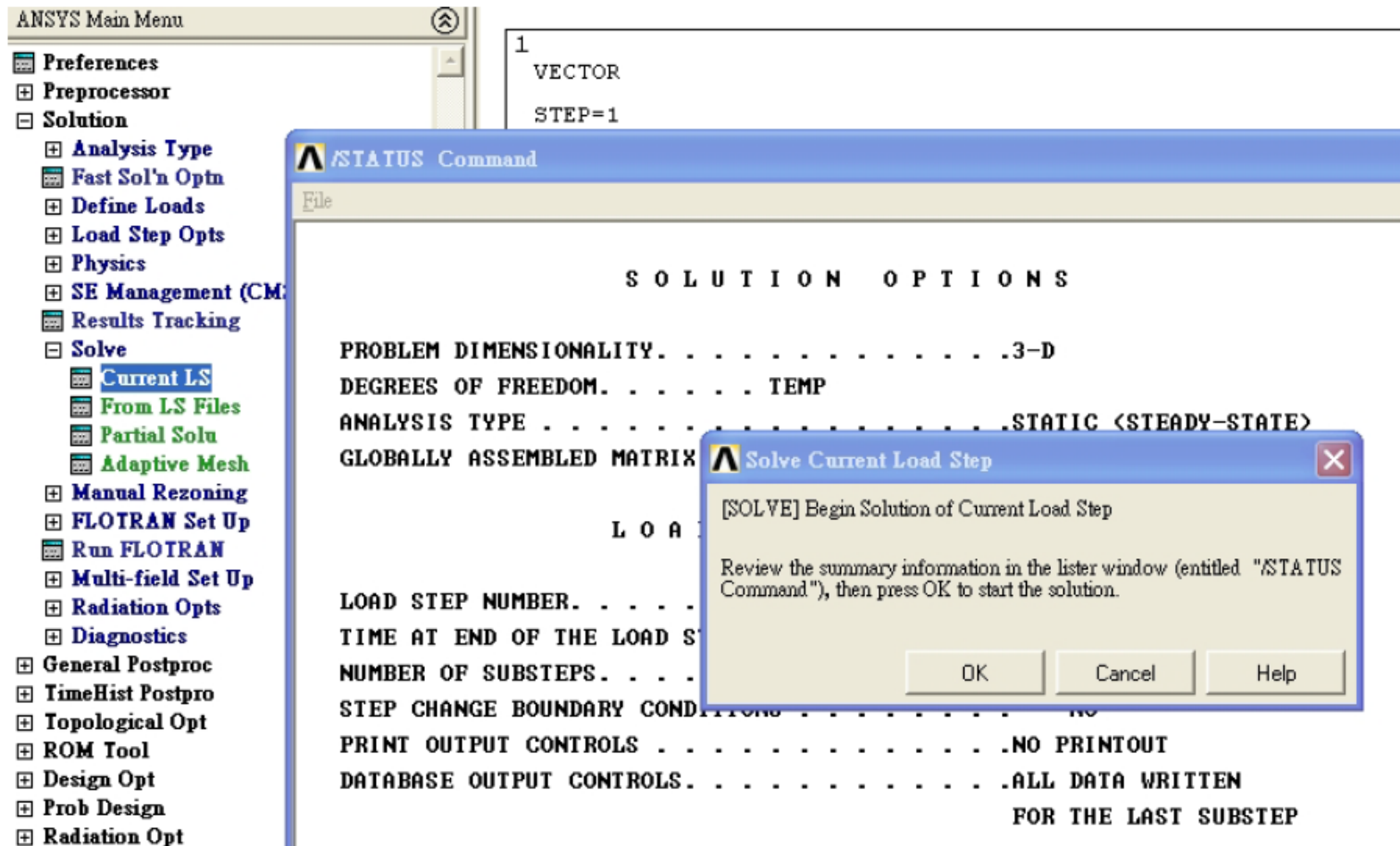
$$\begin{aligned}\text{晶片體積} &= 0.02 * 0.02 * 0.002 \\ &= 8 * 10^{-7} \text{ m}^3\end{aligned}$$

$$\begin{aligned}\text{Heat generation on chip volume} \\ &= (6 \text{ W}) / (8 * 10^{-7} \text{ m}^3) \\ &= 7.5 * 10^6 \text{ W / m}^3\end{aligned}$$

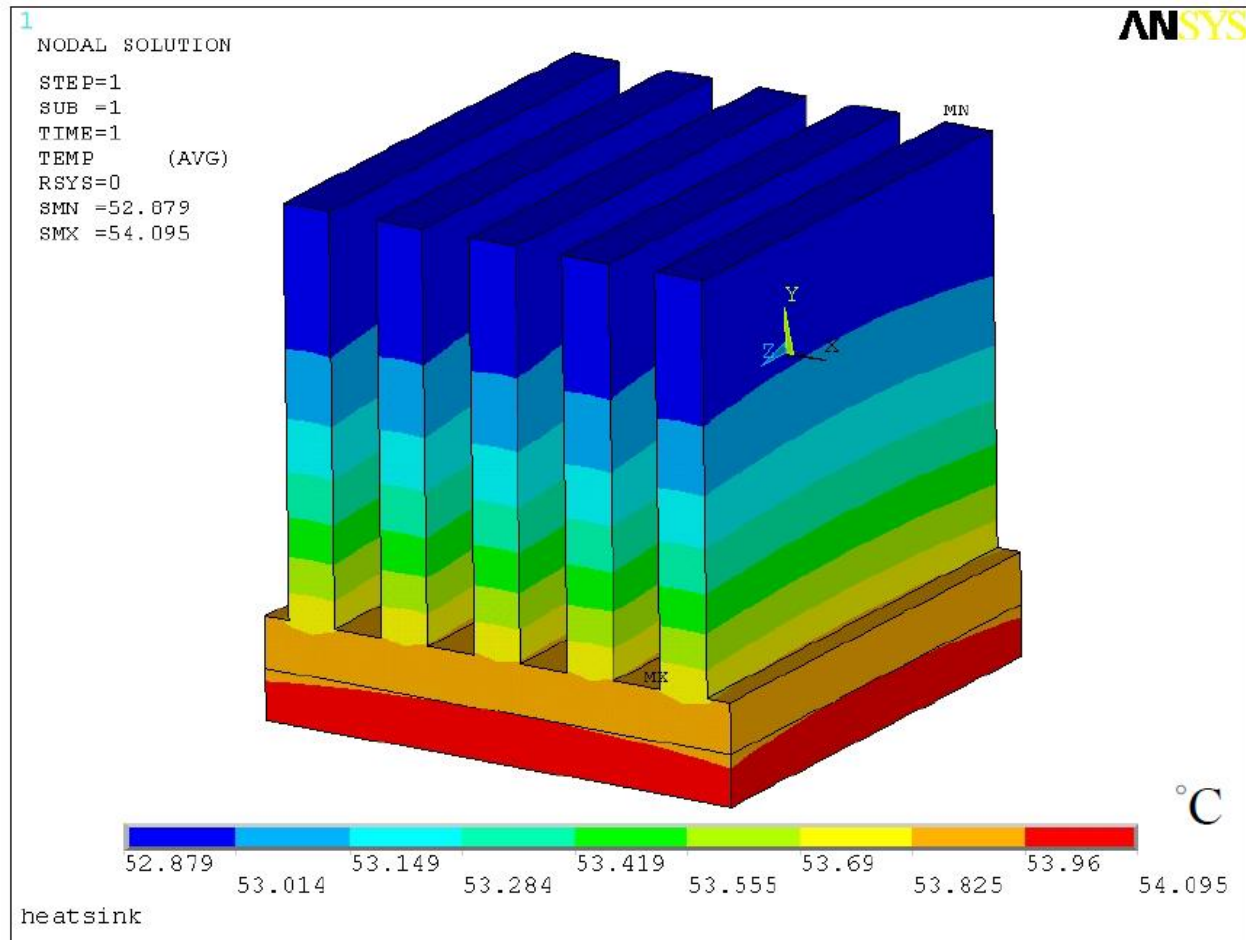
此值輸入ANSYS



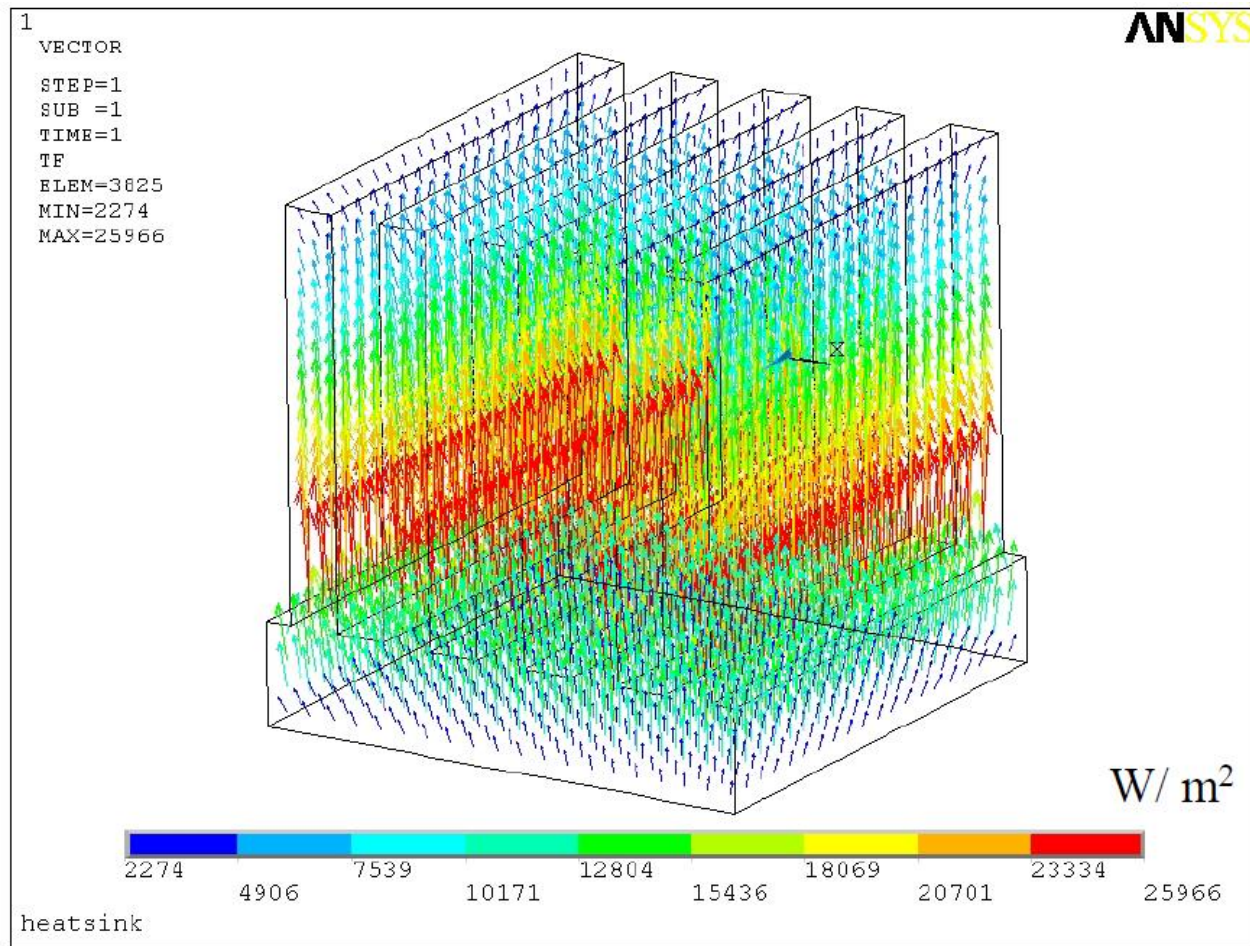
求解



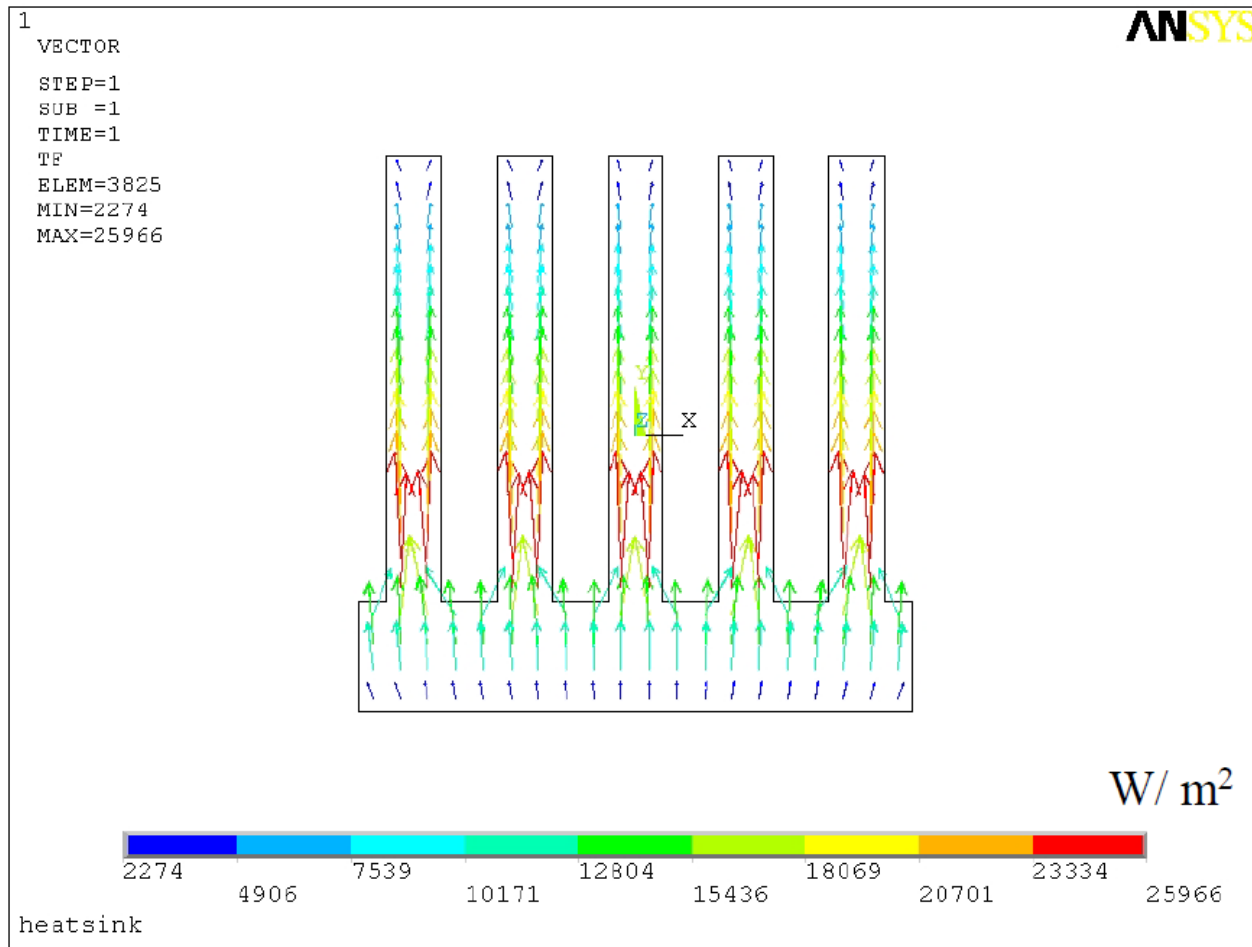
後處理

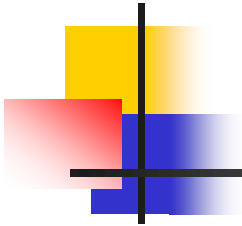


後處理(續)



後處理(續)



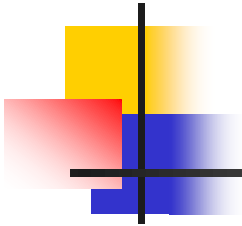


討論

□ Chip 功率

□ 對流係數 h

□ CFD (ANSYS Flotran)



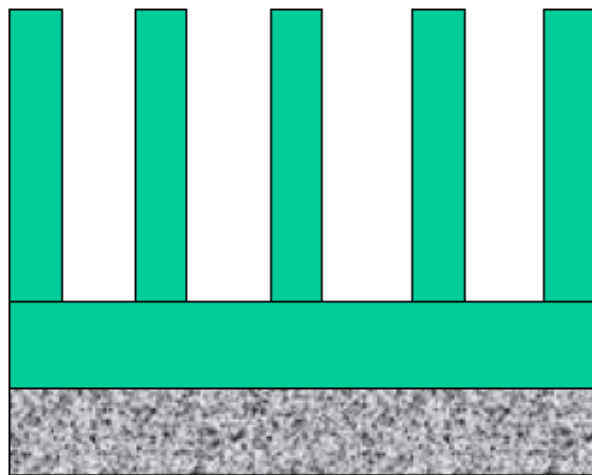
練習

☐ 將溫度單位由 $^{\circ}\text{C}$ 改為 K , 再分析一次.

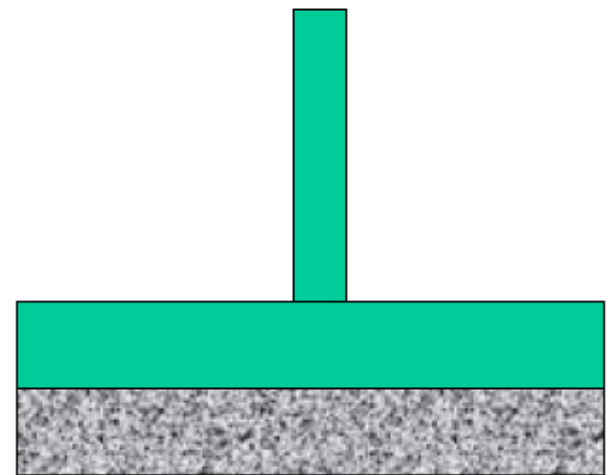
☐ 換算: $(\quad)\text{K} = (\quad)^{\circ}\text{C} + 273.15$

練習

□ 比較散熱效果



chip



chip